5. ARMA(1, 2) process

$$X_t = \phi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

$$\left[\text{Seek } \phi(B) X_t = \theta(B) Z_t \right.$$
$$\left. \Rightarrow X_t = \psi(B) Z_t \right]$$

Here:

$$\underbrace{(1 - \phi_1 B)}_{\phi(B)} X_t = \underbrace{(1 + \theta_1 B + \theta_2 B^2)}_{\theta(B)} Z_t$$

$$\Rightarrow X_t = \frac{(1 + \theta_1 B + \theta_2 B^2)}{1 - \phi_1 B} Z_t$$

From 1(a)

$$g_X(\omega) = \frac{\sigma_Z^2}{2\pi} \frac{\theta(e^{i\omega})\ \theta(e^{-i\omega})}{\phi(e^{i\omega})\ \phi(e^{-i\omega})}$$

$$\therefore g_X(\omega) = \frac{\sigma_Z^2}{2\pi} \frac{(1 + \theta_1 e^{i\omega} + \theta_2 e^{i2\omega})(1 + \theta_1 e^{-i\omega} + \theta_2 e^{-i2\omega})}{(1 - \phi_1 e^{i\omega})(1 - \phi_1 e^{-i\omega})}$$

$$= \frac{\sigma_Z^2}{2\pi} \frac{(1 + \theta_1^2 + \theta_2^2 + (\theta_1 + \theta_1\theta_2)(e^{i\omega} + e^{-i\omega}) + \theta_2(e^{i2\omega} + e^{-i2\omega}}{(1 + \phi_1^2 - \phi_1(e^{i\omega} + e^{-i\omega}))}$$

$$= \frac{\sigma_Z^2}{2\pi} \frac{(1+\theta_1^2+\theta_2^2+2\theta_1(1+\theta_2)\cos\omega + 2\theta_2\cos 2\omega)}{1+\phi_1^2 - 2\phi_1\cos\omega}$$

$\omega \in (-\pi, \pi]$

7(a) (i) $X_t = Z_t + \theta Z_{t-1}$

$= \psi(B) Z_t$ where $\psi(B) = 1 + \theta B$

$$\therefore g_X(\omega) = \frac{\sigma_Z^2}{2\pi} \psi(e^{i\omega})\psi(e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi}(1+\theta e^{i\omega})(1+\theta e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi}(1+\theta^2+\theta(e^{i\omega}+e^{-i\omega}))$$

$$= \frac{\sigma_Z^2}{2\pi}(1+\theta^2+2\theta\cos\omega)$$

(ii) $(1-\rho B)^2 X_t = Z_t$

$$\Rightarrow X_t = \frac{1}{(1-\rho B)^2} Z_t$$

$= \psi(B) Z_t$

where $\psi(B) = \frac{1}{(1-\rho B)^2}$

$$g_X(\omega) = \frac{\sigma_Z^2}{2\pi} \psi(e^{i\omega}) \psi(e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi} \left[\frac{1}{(1-\rho e^{i\omega})^2 (1-\rho e^{-i\omega})^2} \right]$$

$$= \frac{\sigma_Z^2}{2\pi} \left[\frac{1}{(1-\rho e^{i\omega})(1-\rho e^{-i\omega})} \right]^2$$

$$= \frac{\sigma_Z^2}{2\pi} \left[\frac{1}{1+\rho^2 - 2\rho \cos\omega} \right]^2 .$$

(b) $X_t = \frac{2}{3} Y_t + \frac{1}{3} Y_{t-1}$

Consider the mean of $\{X_t\}$

$$\mathbb{E}[X_t] = \frac{2}{3}\mathbb{E}[Y_t] + \frac{1}{3}\mathbb{E}[Y_{t-1}]$$

$$= \mu .$$

Spectral density function of $\{X_t\}$:

$$g_X(\omega) = |h(\omega)|^2 g_Y(\omega)$$

Transfer f'n:

From linear filter

$\cancel{g}$ $h(\omega) = \sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}$

Here $b_0 = \frac{2}{3}$; $b_1 = \frac{1}{3}$; $b_j = 0$ o/w.

$$\therefore \quad h(\omega) = \frac{2}{3} + \frac{1}{3} e^{-i\omega}$$

$$|h(\omega)|^2 = \left(\frac{2}{3} + \frac{1}{3} e^{-i\omega}\right)\left(\frac{2}{3} + \frac{1}{3} e^{i\omega}\right)$$

$$= \frac{5}{9} + \frac{4}{9}\left(e^{i\omega} + e^{-i\omega}\right)$$

$$= \frac{5}{9} + \frac{4}{9} \cos\omega$$

$$g_y(\omega) = \sigma_z^2 (1 - \beta \cos\omega)$$

$$\therefore \quad g_x(\omega) = \sigma_z^2 \left(\frac{5}{9} + \frac{4}{9}\cos\omega\right)\left(1 - \beta\cos\omega\right).$$

8. $g_Y(\omega) = |h(\omega)|^2 g_X(\omega)$

$$X_t = \frac{1}{2}X_{t-1} + Z_t$$

$$\Rightarrow \left(1 - \frac{B}{2}\right)X_t = Z_t$$

$$\Rightarrow X_t = \left(1 - \frac{B}{2}\right)^{-1} Z_t$$

$$= \psi(B) Z_t \qquad \psi(B) = \left(1 - \frac{B}{2}\right)^{-1}$$

$$g_X(\omega) = \frac{\sigma_Z^2}{2\pi} \psi(e^{i\omega})\psi(e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi}\left[\frac{1}{\left(1-\frac{1}{2}e^{i\omega}\right)\left(1-\frac{1}{2}e^{-i\omega}\right)}\right]$$

$$= \frac{\sigma_Z^2}{2\pi}\left[\frac{1}{\frac{5}{4} - \cos\omega}\right] \qquad \omega \in (-\pi, \pi]$$

$$X_t = \sum_{j=-\infty}^{\infty} b_j Y_t \qquad b_0 = b_1 = b_{-1} = \frac{1}{3}.$$
$$b_j = 0 \text{ o/w.}$$

Transfer function:

$$h(\omega) = \sum_{j=-\infty}^{\infty} b_j e^{-i\omega j}$$

$$= \frac{1}{3} + \frac{1}{3}e^{i\omega} + \frac{1}{3}e^{-i\omega}$$

$$= \frac{1}{3}(1 + 2\cos\omega)$$

$$\therefore |h(\omega)|^2 = \left[\frac{1}{3}(1 + 2\cos\omega)\right]^2$$

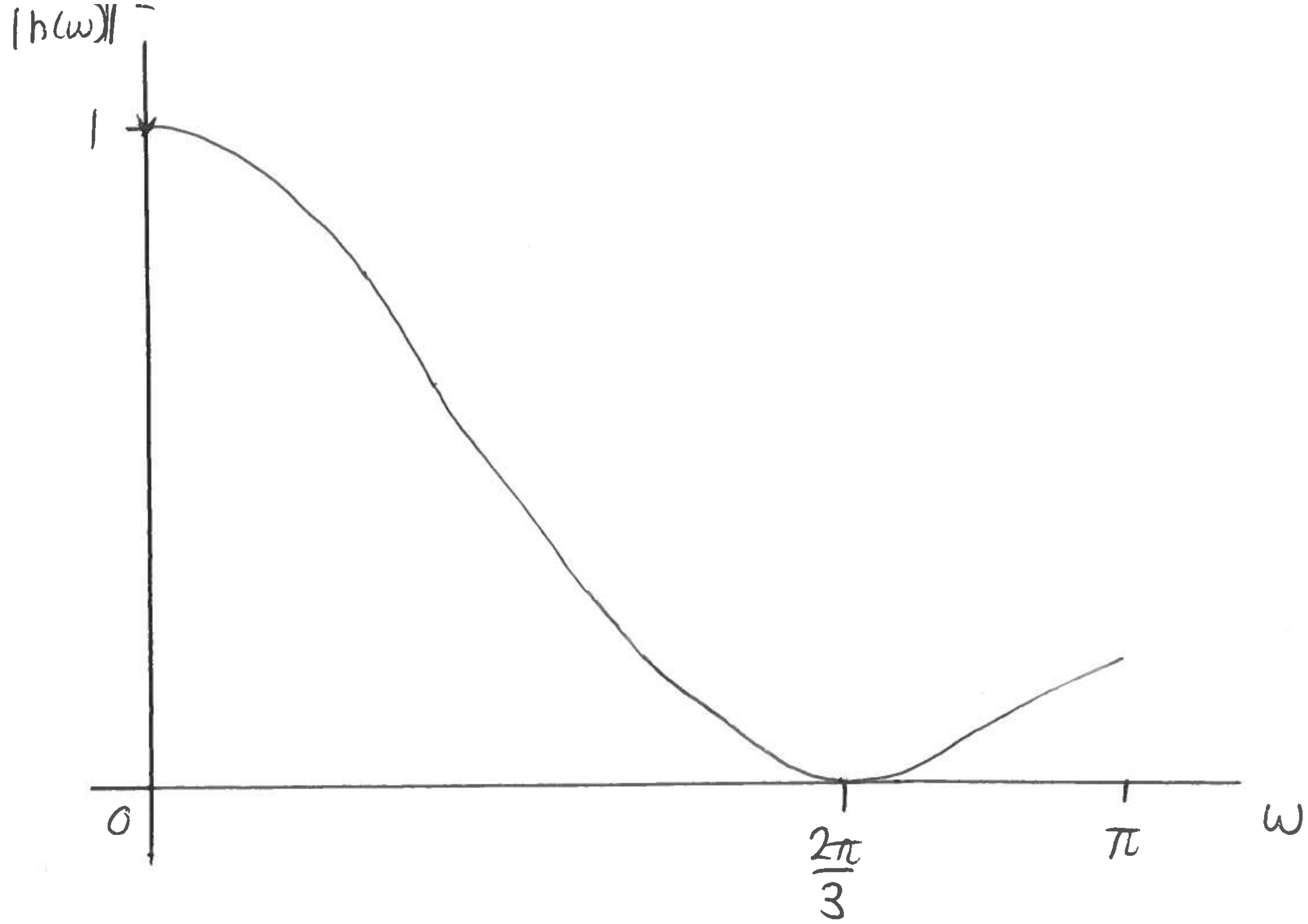
$$= \frac{1}{9}(1 + 2\cos\omega)^2$$

$$g_X(\omega) = |h(\omega)|^2 \, g_Y(\omega)$$

$$= \frac{1}{9}(1 + 2\cos\omega)^2 \times \frac{\sigma_Z^2}{2\pi}\left[\frac{1}{\frac{5}{4} - \cos\omega}\right]$$

$$= \frac{\sigma_Z^2}{9\pi} \frac{2(1 + 2\cos\omega)^2}{(5 - 4\cos\omega)} \qquad \omega \in (-\pi, \pi]$$

(b) $|h(\omega)|^2 = \frac{1}{9}(1 + 2\cos\omega)^2$



|h(ω)|
1
0
2π/3
π
ω